Eric Xu

Curriculum Vitae | *November 2026*

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Research Interests

Scientific machine learning; rare-event sampling and AI ensemble forecasting of weather extremes; data assimilation; dynamical systems; GPU computing for large AI weather models.

Education

University of California, Santa Cruz — Santa Cruz, CA

M.S. in Scientific Computing and Applied Mathematics, Plan I (thesis) — *Expected June 2027*

- One-year accelerated track. Advisor: Prof. Ashesh Chattopadhyay (TACS Lab).

B.S. in Applied Mathematics — *2024 – 2026*

- Highest Honors in the Major. GPA 3.77/4.00. Dean's Honors (4 quarters).

Research Experience

TACS Lab, UC Santa Cruz — Santa Cruz, CA

Researcher, B.S. and M.S. (Advisor: Prof. Ashesh Chattopadhyay) — *Sep 2025 – Present*

- Adapt AI+RES rare-event sampling (Lancelin et al., 2025) to subseasonal forecasting: AI weather-model ensembles (FourCastNet 3, GenCast) are cloned and pruned by score to sample heat waves and cold outbreaks 3 to 4 weeks ahead.
- Built the batch pipeline (JAX, PyTorch; Slurm on H100 nodes, PBS on NCAR Derecho) that launches ensembles weeks before past US extremes and checks against ERA5 (0.25°) whether they reach the observed event.

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ML Researcher (part-time) — *Jul 2026 – Present*

- Made GenCast's 0.25° JAX executable small enough for the persistent compile cache and added persistent GPU workers, so each forecast case compiles the model once instead of at every leg.
- Benchmarked AI weather models (GenCast, FourCastNet 3, SFNO) as ensemble members on H100 nodes and moved the ensemble to FourCastNet 3, cutting GPU-hours per forecast case.

Reports

Eric Xu, Javier Sanchez, Jiayang Wu, Wilson Xie, Edmund Xu. "Full Four-Dimensional Variational Data Assimilation with a Learned Inverse Observation Operator under Observation Noise." AM 170B Applied Mathematics Capstone report, UC Santa Cruz, May 2026. PDF

Selected Projects

4D-Var Data Assimilation with a Learned Inverse Operator | *PyTorch, L-BFGS* — Apr – May 2026

- AM 170B capstone (team of 5): reimplemented Frerix et al. (ICML 2021) in PyTorch and extended it with the full 4D-Var cost, Gaussian observation noise at three levels, and sliding-window cycling, on Lorenz-96 and 2D Kolmogorov flow.
- On Lorenz-96, learned-inverse initialization with hybrid optimization gave the lowest analysis RMSE at every noise level.
- Code: github.com/ericxu1070/invobs-data-assimilation | Report: paper/main.pdf

Cross-Species Transfer Learning with DeepSEA | *PyTorch, scikit-learn* — Jun 2026

- Fine-tuned the human-trained DeepSEA CNN on mouse ENCODE DNA windows. It beat training from scratch on 115 of 133 chromatin features (median AUROC 0.877 vs 0.865) and converged in fewer epochs.
- Showed that zero-shot transfer to mouse scores near chance (median AUROC 0.50), so the model needs fine-tuning.
- Code: github.com/ericxu1070/DeepSEA-Transfer-Learning

Council of Sumer | *Python, FastAPI, AGNTCY SLIM, OpenTelemetry* — Oct 2026

- Built a council of five LLM agents with conflicting goals (hackathon, team of 3): they run as separate processes, debate over AGNTCY SLIM, and vote under runoff and human tie-break rules to govern the city in Hamurabi.
- Co-wrote the deliberation engine and benchmark: over 16 seeded games the council survived 11 of 15 completed reigns vs 9 of 16 for a single LLM, and starved 27% fewer people.
- Code: github.com/ericxu1070/council-of-sumer

Teaching

Baskin School of Engineering, UC Santa Cruz Santa Cruz, CA

Teaching Assistant *Fall 2026*

- AM 20 Ordinary Differential Equations for Engineers; AM 11A Mathematical Methods for Economists I.
- Lead four weekly discussion sections (two per course) and office hours, and grade assignments.
- Pedagogy training: AM 200 Teaching in Applied Mathematics (Fall 2026).

Tutor and Reader *Fall 2025 – Spring 2026*

- Tutor for AM 10 Mathematical Methods I (Linear Algebra), 3 consecutive quarters.
- Reader for CSE 16 Applied Discrete Mathematics; tutor for MATH 19B Calculus II.

Honors

Highest Honors in the Major, B.S. Applied Mathematics, UC Santa Cruz, 2026
Dean's Honors, UC Santa Cruz, 4 quarters

Skills

Languages: Python (primary), C++ (working knowledge), MATLAB, R, Bash; familiar: RISC-V assembly
ML and Scientific Computing: PyTorch, JAX, NumPy, scikit-learn; familiar: Triton (GPU kernels)
HPC and Tools: Slurm, PBS, NCAR Derecho, multi-GPU H100 nodes, Linux, Git, SSH
Mathematics: Linear algebra, ODEs and PDEs, dynamical systems, numerical methods, probability and statistics

Coursework

Graduate: Scientific Machine Learning (AM 261); Applied Dynamical Systems (AM 224); Machine Learning and AI in Genomics (AM 234); Numerical Linear Algebra (AM 213A, in progress); Teaching in Applied Mathematics (AM 200, in progress)

Mathematics and numerics: Mathematical Methods I–III (AM 10, 20, 100); Intro to PDEs (AM 112); Dynamical Systems (AM 114); Scientific Computing (AM 129); Computational Methods (AM 147); Mathematical Modeling I–II (AM 170A/B); Applied Discrete Mathematics (CSE 16)

Probability, statistics and finance: Probability and Statistics (CSE 107); Classical and Bayesian Inference (STAT 132); Financial Mathematics (MATH 114)

Computing: GPU Programming (AM 148); Computer Systems and Assembly (CSE 12)

Additional Information

Citizenship: U.S. citizen

References

Available on request.